

DEAKIN UNIVERSITY

FULL STACK DEVELOPMENT: SECURE FRONTEND APPLICATIONS

ONTRACK SUBMISSION

Newsletter

Submitted By:

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2026/09/19 21:08

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Task Outcomes	Supports
TLO1 Demonstrate engagement with the following unit learning outcomes (legacy)	ULO Unit Learning Outcome 1

September 19, 2026



Newsletter

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DEV@Deakin's newsletter form connects the React interface to an Express backend. A visitor enters an email address to request a welcome email; an account or paid membership is not required.

Repository: <https://gitlab.deakin.edu.au/s222528574/task-p5/-/tree/review-p5-20260917/reviewed-app>

Hosted application: <https://sit313-p5-draft--dev-deakin-sit313-romil.netlify.app>

Demonstration video: [P5 Newsletter walkthrough on Panopto](#) (5:02).

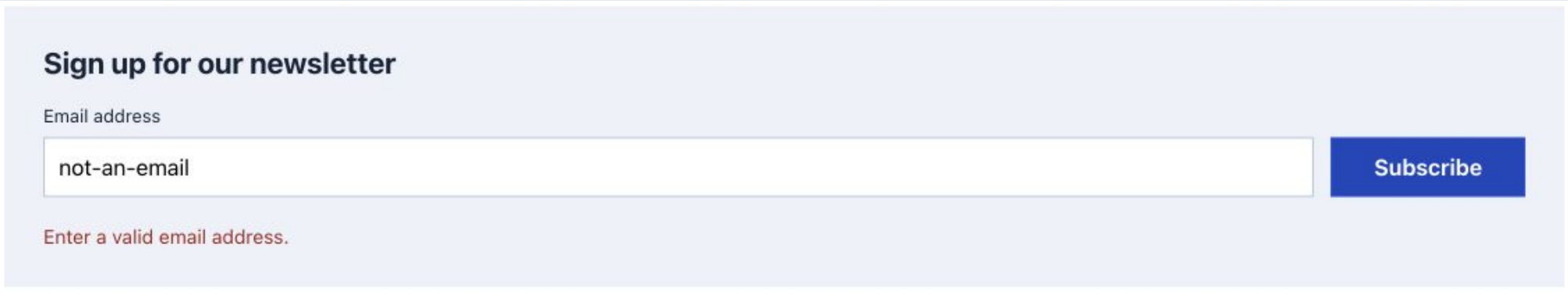
Published source: commit [5ab49f5](#); pipeline [8462 passed](#). The repository is private, with Aaron Spence and Nisha Kumari as Reporters.

React to Express to Gmail

The controlled form holds its email, pending state and feedback in React state. It sends JSON with `fetch` to `POST /api/newsletter`. Express validates the request again, then the server-side Gmail adapter obtains an OAuth access token and submits a MIME welcome message. OAuth credentials remain on the server. Same-origin API routing lets the browser call Express without a separate cross-origin connection.

Input validation

The shared Zod schema trims and lowercases the address, checks its format and limits its length. Invalid input displays an error. While a valid request is pending, the Subscribe button and a submission guard prevent repeated sends.



The screenshot shows a light blue rectangular form titled "Sign up for our newsletter". Below the title, the text "Email address" is displayed. A white input field contains the text "not-an-email". To the right of the input field is a blue button with the text "Subscribe". Below the input field, a red error message reads "Enter a valid email address."

Figure 1. The newsletter form rejects an invalid email address and presents a clear validation message.

Response handling and provider acceptance

Express returns HTTP 202 only after the Gmail adapter confirms acceptance. The interface then clears the input and reports that the welcome email is queued, rather than claiming inbox delivery.

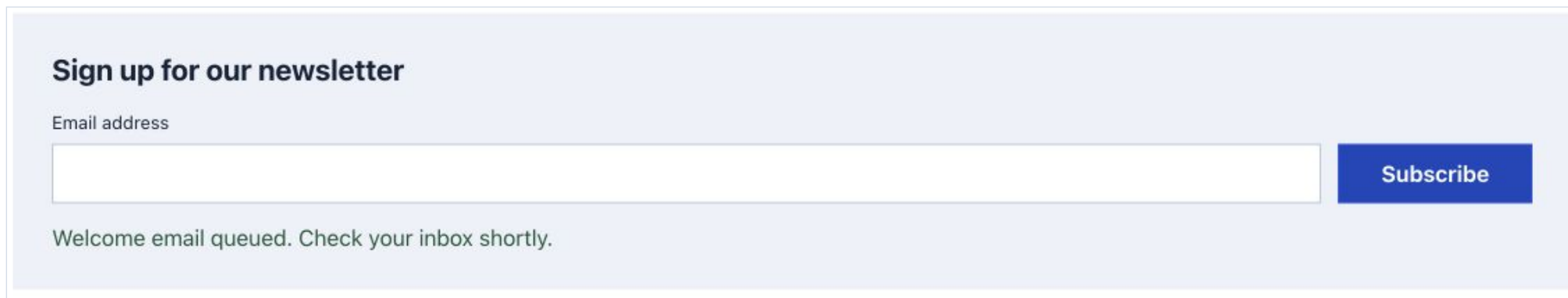
A screenshot of a web form titled "Sign up for our newsletter". It features a text input field labeled "Email address" and a blue "Subscribe" button. Below the input field, a green message states: "Welcome email queued. Check your inbox shortly." The form is set against a light blue background.

Figure 2. Local application feedback after an accepted newsletter request.

Actual Gmail response

```
P5 server listening on http://127.0.0.1:5180.  
{"service":"newsletter","provider":"Gmail","status":200,"outcome":"accepted"}
```

Figure 3. A fresh local request produced Gmail HTTP 200 with outcome accepted in the running P5 server terminal.

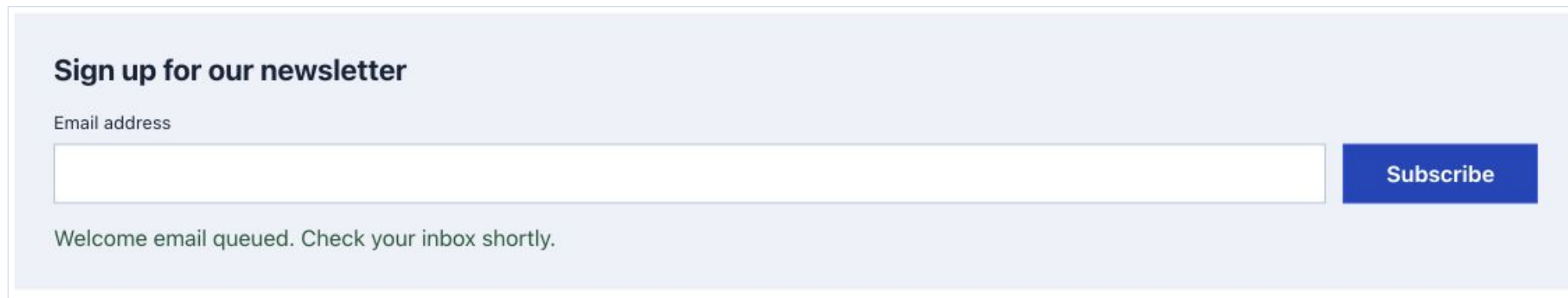
Gmail success requires HTTP 200 and a valid message ID. The console records the service, provider, observed status and outcome, not email contents or credentials. Provider acceptance and recipient delivery are separate checks.

Failure handling

Invalid data returns HTTP 400; a rejected email-send request returns 502; unavailable configuration or an unconfirmed outcome returns 503. Try/catch handling keeps failures visible and preserves the email input. When a result cannot be confirmed, the user is asked to check their inbox before retrying. The server does not automatically repeat the send.

Hosted test and received message

The deployed P5 application also completed a real newsletter request, with Gmail acceptance and the recipient inbox checked separately.



Sign up for our newsletter

Email address

Subscribe

Welcome email queued. Check your inbox shortly.

Figure 4. Queued-success feedback from the hosted P5 application during the 23:25 test on 16 September 2026.

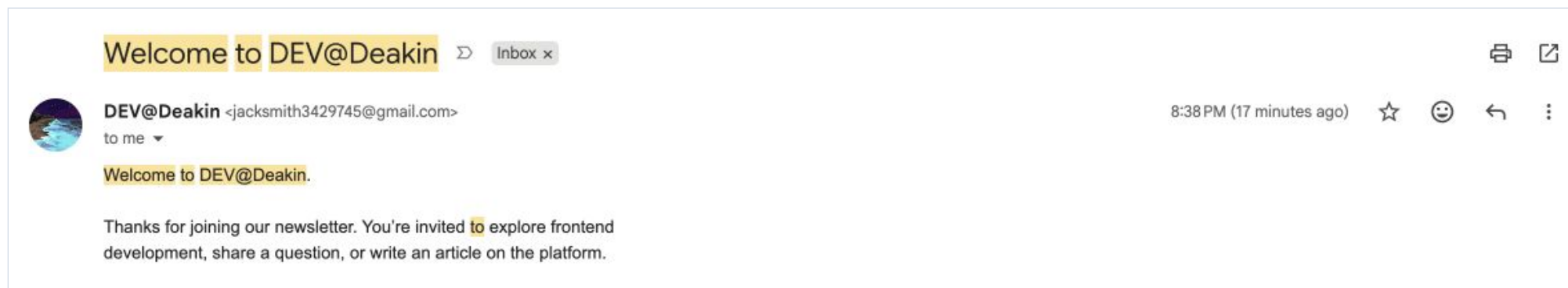


Figure 5. The welcome email received by romilbij@gmail.com at 20:38 on 19 September 2026 during the recording-day test.

Verification. 67 backend and 53 UI checks passed, including validation, request handling, provider failures and pending-state behaviour. Automated provider tests use test doubles; the terminal and inbox evidence above records separate real requests.